BlueQuery - An AI-Powered Conversational System for ARGO Ocean Data Discovery & Visualization

A MINI PROJECT REPORT

GE23627 DESIGN THINKING AND INNOVATION

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BONAFIDE CERTIFICATE

Certified that the mini project titled **“BlueQuery - An AI-Powered Conversational System for ARGO Ocean Data Discovery & Visualization”** is the Bonafide work carried out by **“Kabel B (2116231801076), Karthick S (2116231801079) and Meenakshi G (2116231801099)”** under my supervision. This project has been completed in partial fulfilment of the requirements of the course GE23621 – Design Thinking and Innovation, promoting project-based learning and practical application of design principles.

Certified further that, to the best of my knowledge and belief, the work reported herein is original and does not form part of any other project, report, or work submitted previously for any academic award.

This project contributes toward achieving the following Sustainable Development Goals (SDGs):

1. **SDG 14 — Life Below Water**
2. **SDG 9 — Industry, Innovation and Infrastructure**
3. **SDG 13 — Climate Action**

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ABSTRACT

Oceanographic datasets from the ARGO float program provide critical insights into ocean temperature, salinity, and climate dynamics. However, these datasets are typically stored in complex NetCDF formats, requiring specialized tools and programming expertise to access and analyze. This creates a significant barrier for students, policymakers, and interdisciplinary researchers. This project presents **BlueQuery**, a cloud-native, AI-powered conversational platform designed to simplify access to ARGO ocean data. The system enables users to interact with large-scale oceanographic datasets using natural language queries, eliminating the need for coding or domain-specific tools. The platform integrates a complete data pipeline, including data ingestion from multiple international sources, structured storage in a PostgreSQL database, and an AI-driven query engine that converts user questions into executable SQL queries. It further provides real time visualizations and scientific analysis, including thermocline detection, ocean heat content estimation, and time-series trends. By applying a design thinking approach, this project addresses the gap between data availability and usability, transforming complex ocean data into accessible, interpretable, and actionable insights.

Keywords — ARGO float data, oceanographic data analysis, natural language interface, AI-powered systems, data visualization, thermocline detection, ocean heat content, time-series analysis.

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LIST OF ABBREVIATIONS

Abbreviation	Expanded Form
AI	Artificial Intelligence
API	Application Programming Interface
ARGO	Array for Real-time Geostrophic Oceanography
AWS	Amazon Web Services
BGC	Biogeochemical
ETL	Extract, Transform, Load
GDAC	Global Data Assembly Center
INCOIS	Indian National Centre for Ocean Information Services
LLM	Large Language Model
ML	Machine Learning
NetCDF	Network Common Data Form
NLP	Natural Language Processing
OHC	Ocean Heat Content
PostgreSQL	PostgreSQL
SQL	Structured Query Language

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CHAPTER 1
INTRODUCTION

1.1 Design Thinking Approach

Design Thinking is a structured, human-centered approach to problem-solving that emphasizes understanding user needs, redefining problems, and developing innovative solutions through iterative cycles. Unlike traditional engineering approaches that focus primarily on technical feasibility, Design Thinking integrates **user desirability, technical feasibility, and business viability** to create meaningful and impactful solutions.

In this project, Design Thinking was adopted as the core methodology to address the challenge of accessing and analyzing oceanographic data. Despite the availability of large-scale datasets such as ARGO float data, users often struggle due to complex file formats, lack of intuitive tools, and the need for programming expertise. This gap highlighted the need for a solution that prioritizes usability and accessibility over purely technical implementation.

Several established Design Thinking models guided this work, including the Stanford d.school model, the Double Diamond framework, and the IDEO approach. While these models differ slightly in structure, they share common principles such as understanding user perspectives, clearly defining problems, encouraging creative exploration, building prototypes rapidly, and continuously improving solutions through feedback.

The application of these principles in this project ensured that the solution was not only technically robust but also aligned with real-world user needs. Instead of directly building a data processing system, the focus was first placed on identifying how users interact with ocean data, what challenges they face, and what kind of outputs they expect.

A key aspect of this approach was shifting the perspective from **"how to process data"** to **"how users want to experience data."** This shift led to the idea of a conversational interface, where users can interact with complex datasets using simple natural language rather than technical queries or scripts.

1.2 Stanford Design Thinking Model

The Stanford Design Thinking model consists of five iterative phases: Empathize, Define, Ideate, Prototype, and Test. These phases are not strictly linear but are interconnected, allowing continuous refinement based on insights and feedback.

1. Empathize Phase

In the empathize stage, extensive effort was made to understand the challenges faced by different user groups, including students, researchers, policymakers, and general users. Multiple empathy maps were created to capture user behaviors, thoughts, frustrations, and expectations.

The findings revealed that most users found ARGO data difficult to use due to:

- Complex NetCDF file formats
- Lack of clear documentation
- Dependence on programming tools such as Python or MATLAB
- Difficulty in interpreting scientific outputs

Users expressed a strong need for simplified access, visual insights, and faster interpretation without relying on domain expertise.

2. Define Phase

Based on the insights gathered, the problem was clearly defined to focus on usability rather than data availability. The key issue identified was not the absence of data, but the difficulty in accessing and understanding it.

The final problem statement was framed as: There is a significant gap between ocean data availability and its usability for non-technical users due to complex formats, lack of intuitive interfaces, and dependence on programming knowledge.

This definition helped narrow the scope and ensured that the solution remained focused on user accessibility.

3. Ideate Phase

During the ideation stage, multiple solution ideas were generated through brainstorming sessions and structured discussions. These ideas included interactive dashboards, automated analysis tools, visualization systems, and AI-based interfaces.

After evaluating feasibility and user impact, the concept of an **AI-powered conversational platform** was selected. The idea was to allow users to ask questions in natural language and receive structured outputs, visualizations, and insights automatically.

This phase also led to the identification of key features such as:

- Natural language query system
- Interactive geospatial visualization
- Ocean heat content analysis
- Thermocline detection
- Time-series trend analysis

The ideation process emphasized combining multiple features into a single unified system rather than developing isolated tools.

4. Prototype Phase

In the prototype stage, a working system was developed to demonstrate the feasibility of the solution. The prototype included:

- A data pipeline to process raw ARGO NetCDF files
- A structured database for efficient querying
- A backend API for handling data requests
- An AI module to convert natural language into SQL queries
- A web-based interface for user interaction and visualization

The prototype was designed as an end-to-end system, ensuring that all components—from data ingestion to user interaction—worked seamlessly together.

Special focus was given to making the interface intuitive, with features such as interactive maps, charts, and multiple analysis modules. The goal was to ensure that users could explore ocean data without needing technical knowledge.

5. Test Phase

The system was tested with different types of users to evaluate usability, performance, and accuracy. Feedback was collected on aspects such as ease of use, clarity of results, and response time.

Key observations included:

- Users found the conversational interface intuitive and easy to use
- Visualizations significantly improved understanding
- Some queries required refinement for better accuracy

Based on the feedback, improvements were made to enhance query interpretation, optimize performance, and provide clearer explanations of results.

Application of the Framework

The Design Thinking framework played a crucial role in shaping the development of the project. Each phase contributed directly to the final system:

- Empathy ensured the solution addressed real user problems
- Problem definition kept the project focused and relevant
- Ideation encouraged innovative thinking beyond traditional tools
- Prototyping enabled rapid development and validation
- Testing ensured continuous improvement and usability

As a result, the final system is not just a technical implementation but a user-centered solution that transforms how oceanographic data is accessed and understood.

CHAPTER 2
LITERATURE REVIEW

The Argo Program, established as an international collaborative initiative, has fundamentally transformed global ocean observation. Operating a worldwide array of autonomous profiling floats, the program continuously collects high-resolution measurements of temperature, salinity, and pressure across the upper 2,000 meters of the ocean. These observations are made publicly available through Global Data Assembly Centers (GDACs), representing one of the most comprehensive oceanographic datasets in existence.

Despite this scale of availability, the practical usability of ARGO data remains limited. Datasets are distributed in NetCDF format — a scientific storage standard optimized for multidimensional data — which requires specialized tools, programming expertise, and domain knowledge to decode. Availability and accessibility are not the same thing, and this distinction defines the central problem this project addresses.

2.1 Existing Platforms and Their Limitations

Several platforms have been developed to improve access to ARGO data. The Argovis platform provides an interactive web interface for spatial filtering and basic visualization. Programming libraries such as argopy for Python and argodata for R streamline data access for researchers. INCOIS and the Coriolis Data Centre offer regional portals with download and basic map capabilities.

While these tools represent meaningful progress, they share a common limitation: they are designed for technical users. Argovis still requires users to understand oceanographic parameters and configure queries manually. Argopy and argodata require coding proficiency. None of these systems provide natural language interaction, automated insight generation, or an integrated analysis-to-visualization workflow. The interaction model across all existing platforms remains technical-first and excludes a large potential user base.

2.2 Research in Ocean Data Analysis

Academic research has advanced the analytical side of oceanographic data considerably. Machine learning models have been applied to thermocline detection and anomaly identification. Graph-based representations have improved understanding of float trajectory dynamics. Functional data analysis methods have enhanced ocean heat content estimation. These contributions demonstrate the scientific potential of ARGO data but remain confined to research environments — inaccessible to practitioners without specialized training in statistics and programming.

2.3 AI and Natural Language Interfaces

The emergence of large language models has opened a new design space for scientific data interaction. Natural Language to SQL systems allows users to query databases using plain English, bypassing the need for structured query languages. Domain-specific models such as OceanGPT have demonstrated how AI systems can be oriented toward scientific datasets, producing meaningful interpretations rather than raw outputs. Explainable AI techniques, including SHAP, further improve transparency by surfacing how individual variables contribute to model predictions — an important consideration in scientific applications where the reasoning behind a result matter as much as the result itself.

2.4 Research Gap

A review of existing systems reveals a consistent and unaddressed gap: no platform currently combines natural language querying, automated data processing, scientific computation, interactive visualization, and explainable AI in a single unified system accessible to non-technical users. BlueQuery was designed specifically to fill this gap for Indian Ocean ARGO data.

CHAPTER 3
DOMAIN AREA

The domain of this project is centered on **Oceanographic Data Analysis**, with a specific focus on the study and interpretation of large-scale ocean data collected through the Argo Program. Oceanographic data plays a critical role in understanding the physical, chemical, and biological processes occurring within the ocean, which directly influence global climate systems, marine ecosystems, and weather patterns.

In recent years, the volume and complexity of ocean data have increased significantly due to advancements in sensing technologies and global observation networks. However, the effective utilization of this data remains a challenge due to its scale, structure, and technical complexity. This project addresses these challenges by focusing on making oceanographic data more accessible, interpretable, and actionable.

3.1 ARGO Float System

The Argo Program is an international collaborative initiative that operates a global array of approximately 4,000 autonomous profiling floats distributed across the world's oceans. These floats continuously collect high-resolution vertical profiles of ocean parameters, enabling large-scale and real-time monitoring of ocean conditions.

Figure 3.1: Global Distribution of Argo Floats

Each float follows a standardized operational cycle. It descends to a parking depth of around 1,000 meters and drifts with ocean currents for a period of approximately ten days. It then descends further to about 2,000 meters before ascending toward the surface while recording temperature, salinity, and pressure data across different depth levels. Once it reaches the surface, the collected data is transmitted via satellite to global data centers within a few hours.

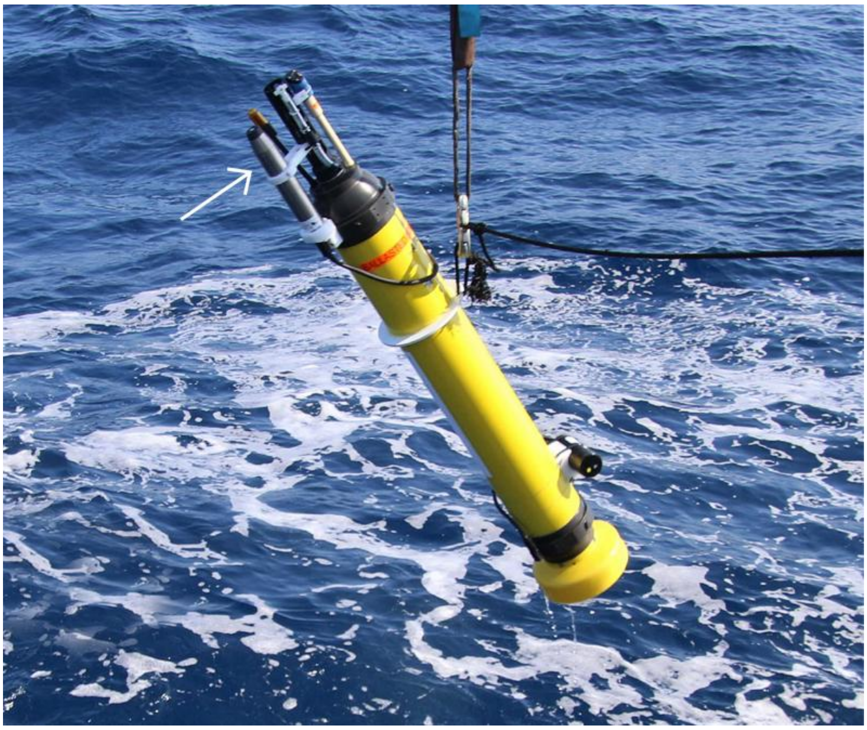


Figure 3.2: ARGO Float

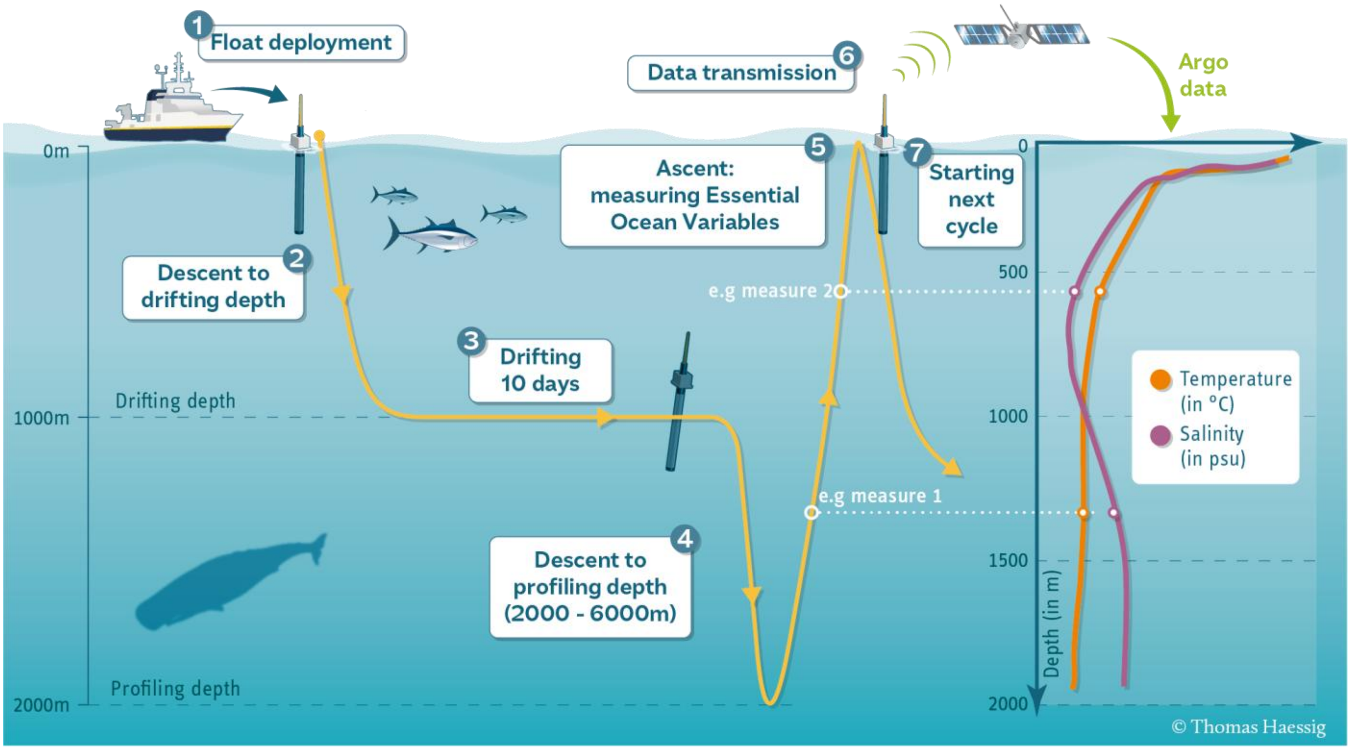


Figure 3.3: Argo Float Measurement Cycle

This continuous cycle enables consistent and wide ocean coverage, including remote regions that are difficult to access through traditional ship-based observations. The

Indian Ocean region, supported by organizations such as INCOIS, plays a significant role in climate monitoring, monsoon prediction, and cyclone analysis.

3.2 Key Parameters Measured

ARGO floats capture a set of core oceanographic parameters that describe both the physical state and spatial context of the ocean. These measurements are essential for analyzing ocean structure, movement, and variability over time.

- Temperature represents the thermal condition of seawater and is used to study heat distribution, seasonal variation, and ocean stratification. It plays a major role in climate analysis and ocean heat content estimation.
- Salinity measures the concentration of dissolved salts in water. It influences density, buoyancy, and large-scale circulation patterns such as thermohaline circulation.
- Pressure (Depth) indicates the vertical position of measurements and is used to reconstruct the water column, helping identify layers like the mixed layer and thermocline.
- Latitude and Longitude provide the geographical location of each observation, enabling spatial analysis of ocean conditions and mapping of float movement across regions.
- Timestamp records the exact time of data collection, allowing temporal analysis such as trends, seasonal patterns, and time-series studies.

In addition to these core parameters, some floats are equipped with biogeochemical (BGC) sensors that measure variables such as dissolved oxygen, pH, nitrate, and chlorophyll. However, these parameters are not available for all floats and are treated as optional (nullable) fields in the dataset, depending on float type and sensor availability.

3.3 Characteristics of Oceanographic Data

Oceanographic data collected from ARGO floats has unique characteristics that make it both valuable and challenging to process.

- **Multidimensional Structure:** Data exists across multiple dimensions, including depth, time, latitude, and longitude, requiring specialized tools for analysis.

- **Large Scale:** Millions of observations are generated continuously, resulting in massive datasets that grow over time.
- **Heterogeneous Sources:** Data is collected and maintained by multiple international data centers such as INCOIS, Coriolis, and CSIRO, leading to variations in schema and format.
- **Complex File Formats:** Data is stored in NetCDF format, which requires dedicated libraries and preprocessing before analysis.
- **Quality Control Flags:** Each data point includes quality indicators that must be interpreted correctly to ensure accurate analysis.

These characteristics increase the complexity of working with oceanographic data, especially for users without technical expertise.

3.4 Challenges in the Domain

Despite its importance, oceanographic data analysis faces several challenges that limit its usability:

- **Data Accessibility Issues:** Although data is publicly available, it is not easily usable without technical tools
- **High Learning Curve:** Requires knowledge of programming languages and scientific concepts
- **Lack of Integrated Systems:** Data processing, analysis, and visualization are often handled separately
- **Interpretation Complexity:** Raw data does not directly provide insights and requires further analysis
- **Limited User-Friendly Interfaces:** Existing tools are designed primarily for experts rather than general users

These challenges create a gap between data availability and meaningful usage.

3.5 Scientific and Societal Significance

Oceanographic data has wide-ranging applications that are critical for both scientific research and societal needs. The ability to analyze ocean data effectively enables better understanding and decision-making in multiple domains.

- **Climate Monitoring:** Oceans absorb a significant portion of global heat. Temperature and salinity data from ARGO floats are essential for studying global warming, ocean heat content, and long-term climate variability.
- **Cyclone and Weather Forecasting:** Ocean surface temperature and subsurface heat content influence cyclone formation and intensity. Accurate ocean data improves weather prediction models.
- **Monsoon Prediction:** Variations in ocean conditions, particularly in the Indian Ocean, affect monsoon patterns, making ARGO data vital for seasonal forecasting.
- **Marine Ecosystem Analysis:** Parameters such as temperature, salinity, and oxygen levels help in understanding marine biodiversity, productivity, and ecological balance.
- **Fisheries and Coastal Management:** Ocean conditions determine fish distribution and habitat suitability, supporting sustainable fisheries and coastal planning.

3.6 Challenges in Oceanographic Data Analysis

Despite its importance, several challenges limit the effective use of oceanographic data.

- **Limited Accessibility:** Although data is publicly available, it is not easily usable without technical tools and domain knowledge.
- **High Learning Curve:** Users must be familiar with programming languages and scientific data formats to perform analysis.
- **Fragmented Systems:** Data retrieval, processing, analysis, and visualization are often handled using separate tools, increasing complexity.
- **Interpretation Difficulty:** Raw data does not directly provide insights and requires additional processing and analysis.

- **Lack of User-Friendly Interfaces:** Existing platforms are primarily designed for experts, making them difficult for general users.

These challenges create a gap between the availability of data and its practical usability.

3.7 Scope of the Domain in This Project

This project focuses on addressing the gap between complex oceanographic data and user-friendly interaction. The domain is approached from both a technical and usability perspective to ensure that data can be accessed and interpreted effectively by a wide range of users.

The key focus areas include:

- Simplifying access to ARGO ocean data
- Enabling natural language-based interaction with datasets
- Providing real-time visualization and analysis
- Integrating scientific computations such as thermocline detection and ocean heat content
- Supporting both technical and non-technical users

By combining data engineering, artificial intelligence, and visualization techniques, the project aims to transform complex ocean data into meaningful and actionable insights.

3.8 Domain Relevance to Modern Applications

Oceanographic data has become increasingly important in addressing global challenges such as climate change, disaster management, and environmental sustainability. Governments, research institutions, and international organizations rely on accurate and timely ocean data for policy-making and strategic planning.

The integration of modern technologies such as cloud computing, artificial intelligence, and interactive visualization enhances the usability and impact of this data. These technologies enable efficient data processing, intelligent analysis, and intuitive interaction, making it possible to convert raw datasets into valuable insights for real-world applications.

CHAPTER 4
EMPATHIZE STAGE

The empathize phase was conducted to develop a grounded understanding of how different users engage with oceanographic data — their workflows, their frustrations, and their unmet needs. Research involved structured observation, informal interviews, and the construction of detailed empathy maps for ten individual users across eight distinct user groups.

4.1 User Groups Identified

The following user groups were studied, selected to represent the full spectrum of potential users from highly technical to entirely non-technical backgrounds:

- Marine researchers and oceanographers
- Climate scientists and analysts
- Students across engineering, biotechnology, and environmental sciences
- Assistant professors and educators
- Environmental and policy analysts
- Robotics and systems engineers working with marine platforms
- Navy and maritime professionals
- General users relying on secondary sources for ocean information

4.2 Empathy Mapping Process

Ten individual empathy maps were constructed, each covering four dimensions of user experience: what the user sees in their working environment, what they hear from colleagues and systems, what they think and feel about the data interaction process, and what they say and do when attempting to access ocean data. Pain points and gains were recorded separately for each user.

The following structured questions were used during the empathy mapping process to better understand user behavior, emotions, workflows, and expectations regarding oceanographic data interaction.

Think and Feel

1. How confident do you feel working with complex data formats like NetCDF without technical support?
2. What frustrates you most when analyzing large oceanographic datasets?
3. How do you feel when data analysis is delayed due to complex tools or lack of expertise?
4. How important is it for you to understand ocean parameters without deep coding knowledge?

Hear

1. What do you usually hear from colleagues, reports, or tools about accessing or using ocean data?
2. What advice or warnings do people commonly give you about working with ocean or ARGO data?
3. What do official dashboards, reports, or data portals communicate to you about ocean conditions or trends?

See

1. What do you observe about the accessibility and usability of existing ocean data tools?
2. What difficulties do you observe when non-technical users try to analyze ocean data?
3. How do you see policymakers or educators struggling to interpret scientific ocean data?

Say and Do

1. What do you usually say when explaining your ocean-data needs or challenges to others?

2. When discussing ocean data, what do you often say is missing, unclear, or hard to interpret?
3. What tools or methods do you rely on most when exploring ocean conditions or trends?
4. When time is limited, what shortcuts or workarounds do you use to get ocean information?

Target Users

Marine researchers, Climate scientists, Students, Policy analysts, Educators, Environmental researchers, Oceanographers, Navy / Marines

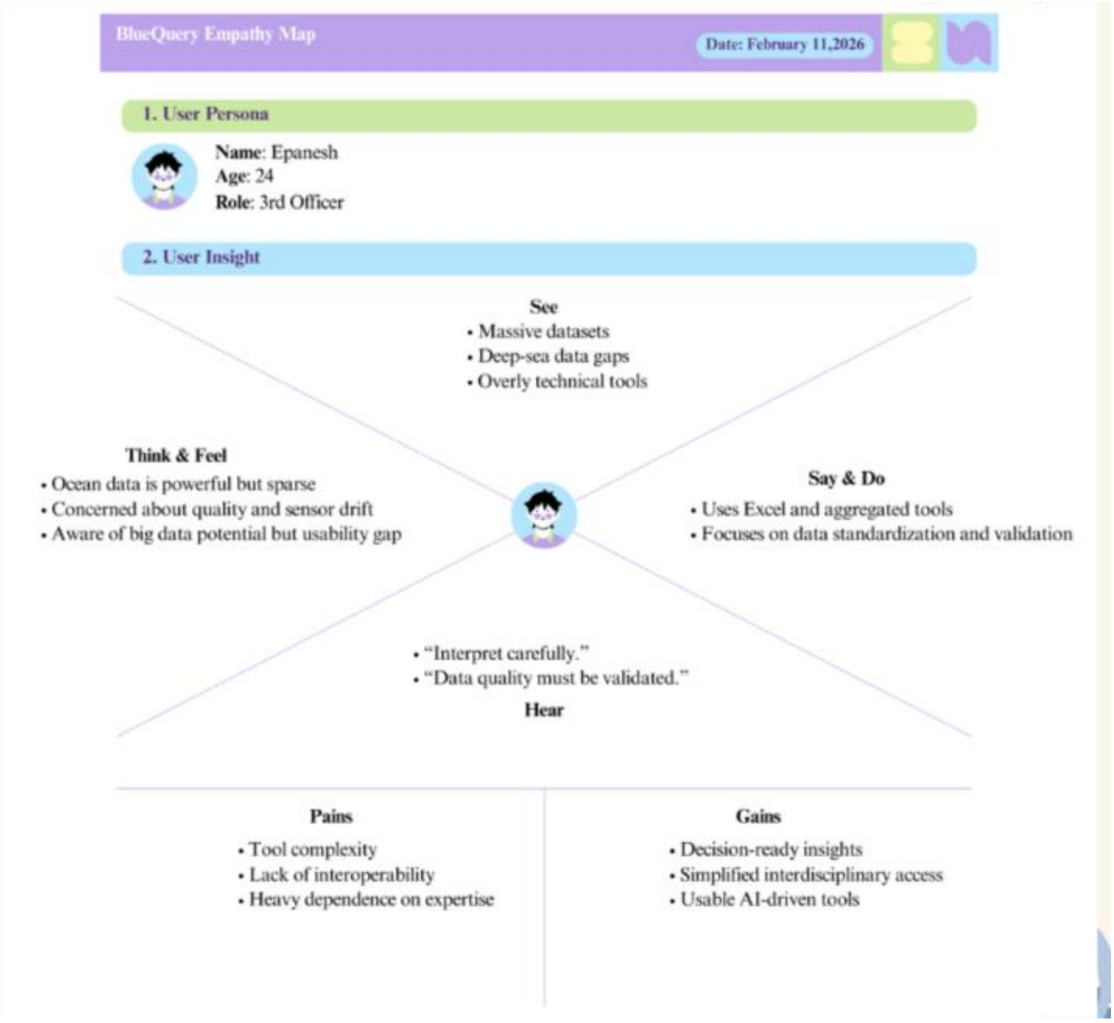


Figure 4.2: Sample Empathy Map

4.3 Key Observations

Across all ten user profiles, a consistent pattern emerged: users do not interact with raw ARGO data directly. They rely instead on pre-processed reports, dashboards, or aggregated summaries produced by domain experts. Even technically proficient users — those who use Python or R — reported spending disproportionate time on data cleaning, schema normalization, and format conversion before any meaningful analysis could begin.

Non-technical users expressed near-complete dependency on experts for interpretation, creating operational delays and reducing the autonomy of decision-makers who need ocean data to inform policy, safety, or resource management. Students with domain interest but limited programming background found existing platforms discouraging rather than enabling.

A marine biology researcher noted reliance on pre-processed climatologies because building custom extraction pipelines from raw NetCDF is too time-consuming for routine work. A climate policy advisor emphasized the need for executive summaries and risk-framed outputs rather than statistical tables. A Navy cadet indicated that while the data is scientifically rich, its presentation does not translate into operationally useful information without significant expert mediation.

4.4 Pain Points Identified

The following pain points were consistently identified across all user groups:

- Complex NetCDF file formats requiring specialized tools and schema knowledge
- High dependency on programming languages such as Python, MATLAB, or R
- Fragmented workflows requiring separate tools for extraction, analysis, and visualization
- Long preprocessing time before any meaningful analysis can begin
- Absence of contextual explanation alongside raw numerical outputs
- Lack of a unified platform integrating data access, analysis, and visualization

- Limited and inconsistent documentation making independent exploration difficult

Table 4.4: Consolidated Pain and Gain Matrix from all ten empathy maps

Pain	Gains
Complex data formats require coding	Simple, user-friendly access
Tools are difficult for beginners	Easy-to-use interface
Data lacks clear explanation	Clear, understandable insights
Information is scattered across platforms	One unified platform
Manual preprocessing takes time	Automated data processing
Visualizations lack context	Interactive charts with explanations
Heavy reliance on experts	Independent user access
Hard to convert data into action	Decision-ready summaries

4.5 User Needs Identified

The empathy phase revealed the following core user needs that any effective solution must address:

- Simple, direct data access without format or tool dependencies
- Visual outputs that communicate patterns without requiring domain expertise to interpret
- Fast response — results within seconds, not hours
- Natural language interaction requiring no coding knowledge
- Contextual explanations that help users understand what the data means scientifically
- A single integrated platform covering the full workflow from query to insight

CHAPTER 5
DEFINE STAGE

5.1 Problem Statement

Based on synthesis of the empathy phase findings, the core problem was defined as follows:

There is a fundamental disconnect between the availability of ARGO oceanographic data and its usability. Complex data formats, the absence of intuitive query mechanisms, high dependence on programming expertise, and fragmented analytical workflows collectively prevent a large population of potential users — including students, researchers, policymakers, and maritime professionals — from extracting value from one of the world's most important environmental datasets.

5.2 Problem Analysis

The problem is not a data scarcity issue. ARGO data is publicly available, continuously updated, and of high scientific quality. The barrier exists entirely at the interaction and usability layer.

NetCDF format imposes a preprocessing requirement before any analysis can begin, immediately excluding users without Python or MATLAB proficiency. Among those who do possess programming skills, significant time is still lost to schema normalization across heterogeneous data sources, quality flag interpretation, and the construction of custom visualization pipelines.

Existing platforms address parts of this problem — Argovis improves spatial accessibility, argopy reduces Python overhead — but none eliminate the technical barrier entirely, and none provide automated scientific interpretation. The workflow remains fragmented: a user must move between a download portal, a processing environment, and a separate visualization tool to complete a single analytical task.

5.3 Design Direction

To address the identified problem, the solution focuses on improving usability while maintaining analytical depth. The key design directions include:

- Enabling natural language-based data querying
- Reducing preprocessing effort through automated data handling
- Integrating data access, analysis, and visualization into a single system
- Providing clear and interpretable outputs for both technical and non-technical users

CHAPTER 6
IDEATION STAGE

The Ideation stage focuses on generating a wide range of possible solutions based on the problem defined in the previous stage. This phase encourages creative thinking and exploration of multiple approaches to address the identified challenges. The objective is to identify innovative solutions that align with user needs while remaining technically feasible.

In this project, brainstorming sessions were conducted to explore different ways to simplify access to oceanographic data and improve user interaction.

6.1 Brainstorming Process

Table 6.1: Ideate Table

Round	**Member 1**	**Member 2**	**Member 3**
1	Interactive ARGO Float Map	Depth Profile Comparison Tool	Marine Ecosystem Health Indicator
2	Ocean Heat Content Indicator	Ocean Trend Analysis	Region-Based Data Filtering
3	AI Insight Summary	Salinity Analysis	Float Trajectory Visualization
4	Thermocline Depth Identification	Hypoxia Alert System	Mixed Layer Depth Detection
5	Stratification Index	Time Series Analysis	Automated Ocean Alert System

6.2 Idea Evaluation and Selection

Dot voting was used to evaluate each idea across three criteria: user impact, technical feasibility, and alignment with the defined problem. Each team member allocated votes across the fifteen ideas independently.

Ideas receiving maximum votes (3/3): Interactive ARGO Float Map, Ocean Heat Content Indicator, Ocean Trend Analysis, Thermocline Depth Identification, Automated Ocean Alert System.

Ideas receiving strong votes (2/3): Depth Profile Comparison Tool, AI Insight Summary, Salinity Analysis, Time Series Analysis.

The following nine features were selected for development, representing a balanced combination of visualization capability, scientific depth, and AI-driven interaction:

1. Interactive ARGO Float Map
2. Ocean Heat Content Indicator
3. AI Insight Summary — Natural Language Query Engine
4. Ocean Trend Analysis
5. Depth Profile Comparison Tool
6. Thermocline Depth Identification
7. Automated Ocean Alert System
8. Salinity Analysis
9. Time Series Analysis

6.3 Value Proposition

"Enable anyone to explore ocean data through simple natural language questions and receive instant, scientifically grounded visual insights — without writing code or possessing domain expertise."

6.4 Proposed System Architecture

A high-level system architecture was designed during the ideation stage to validate the feasibility of the proposed solution and guide development.

The system follows a four-layer modular architecture:

Data Layer → Database Layer → AI Layer → Interface Layer

Layer Overview

- Data Layer: Collects and preprocesses ARGO float data (NetCDF format).
- Database Layer: Stores structured data with indexing for efficient querying.
- AI Layer: Converts natural language queries into database queries and performs analysis.
- Interface Layer: Provides a web-based dashboard with visualizations and user interaction.

Purpose of the Architecture

- Ensures scalability for large datasets
- Maintains modular design
- Supports AI-based querying
- Enables interactive data visualization

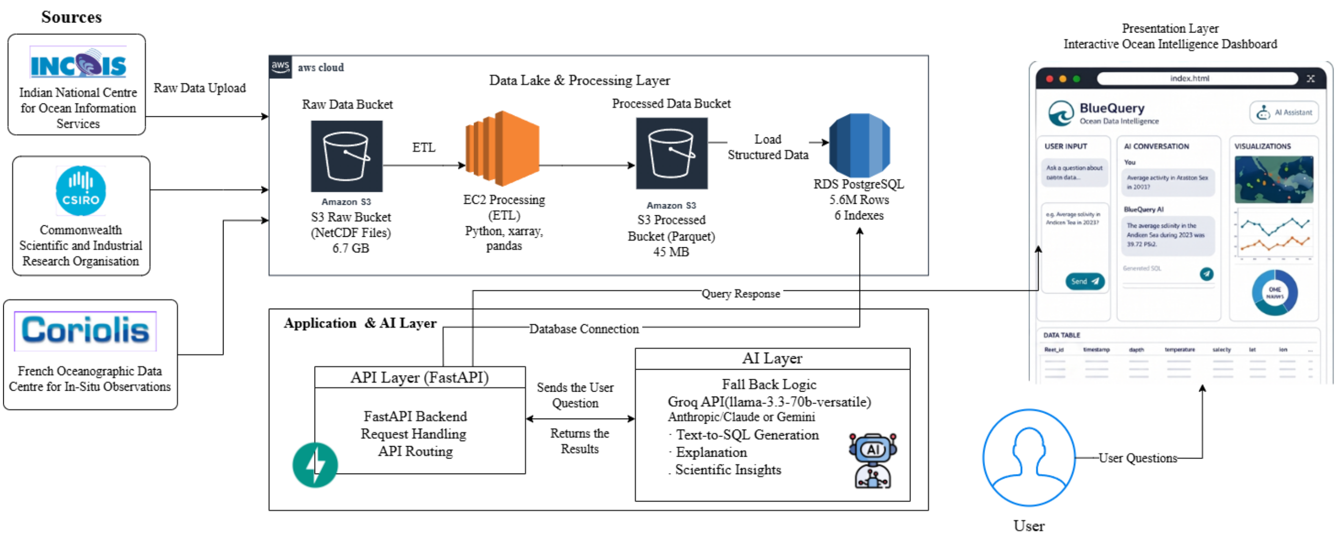


Figure 6.4: System Architecture

CHAPTER 7

PROTOTYPE STAGE

7.1 System Architecture

The system was designed using a modular four-layer architecture consisting of the Data Layer, Database Layer, AI Layer, and Interface Layer. This structured approach ensures clear separation of responsibilities, improves maintainability, and enables independent optimization of each component. The architecture supports efficient data flow from raw data ingestion to end-user interaction, allowing the system to scale while maintaining performance and usability.

7.2 Data Layer

The Data Layer is responsible for collecting and preprocessing raw oceanographic data. ARGO float data in NetCDF format was obtained from multiple international sources and spans several years of observations across the Indian Ocean. Due to the multidimensional and complex nature of NetCDF files, a data processing pipeline was developed to extract key parameters such as temperature, salinity, depth, and geographical coordinates.

The preprocessing stage includes data cleaning, handling missing values, and applying quality control filtering to ensure reliability. Data from different sources is standardized into a consistent schema, enabling seamless integration. The processed data is then converted into a structured format suitable for efficient storage and querying, significantly reducing the complexity of handling raw scientific datasets.

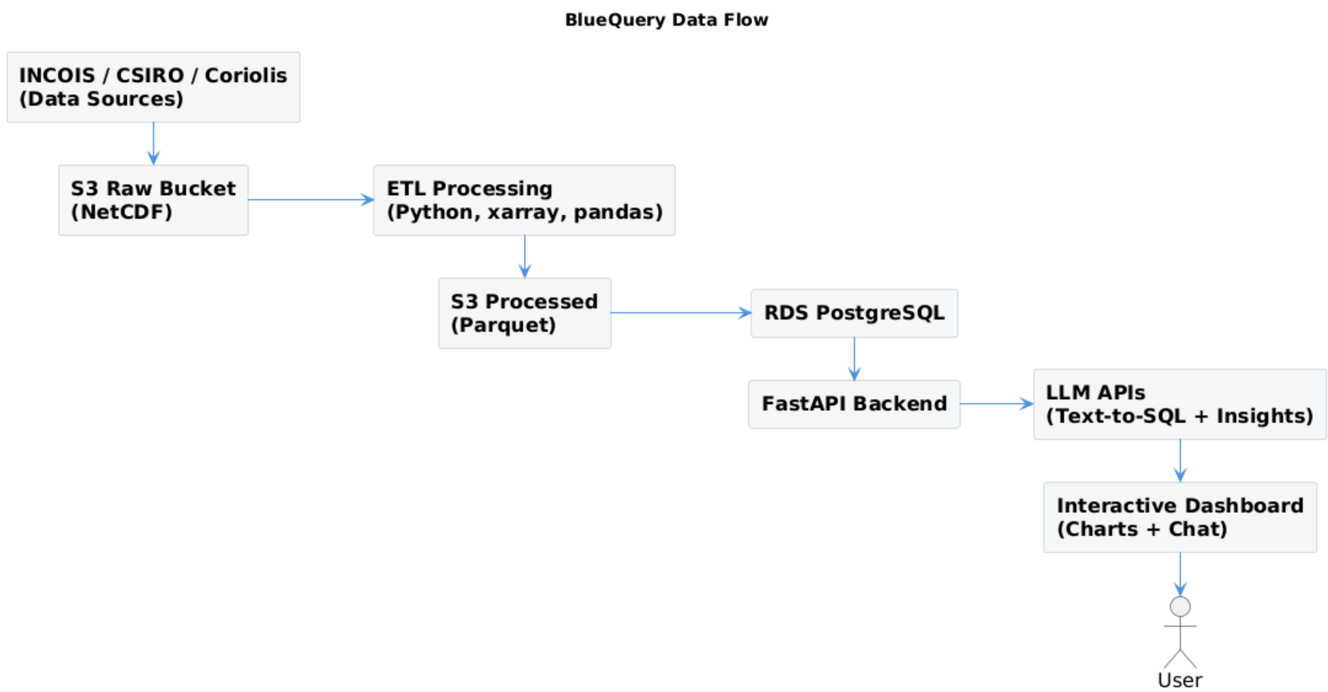


Figure 7.2: Data Pipeline Flow

7.3 Database Layer

The processed data is stored in a PostgreSQL database designed for efficient handling of large-scale structured datasets. The schema includes essential attributes such as float identifiers, timestamps, spatial coordinates, depth levels, temperature, and salinity, with high-precision data types used to maintain scientific accuracy.

To improve performance, indexing strategies are applied on frequently accessed fields including float identifiers, timestamps, spatial attributes, and depth values. These optimizations enable fast execution of queries related to time-series analysis, spatial filtering, and vertical profiling. The database is capable of supporting real-time data access while maintaining consistent response times.

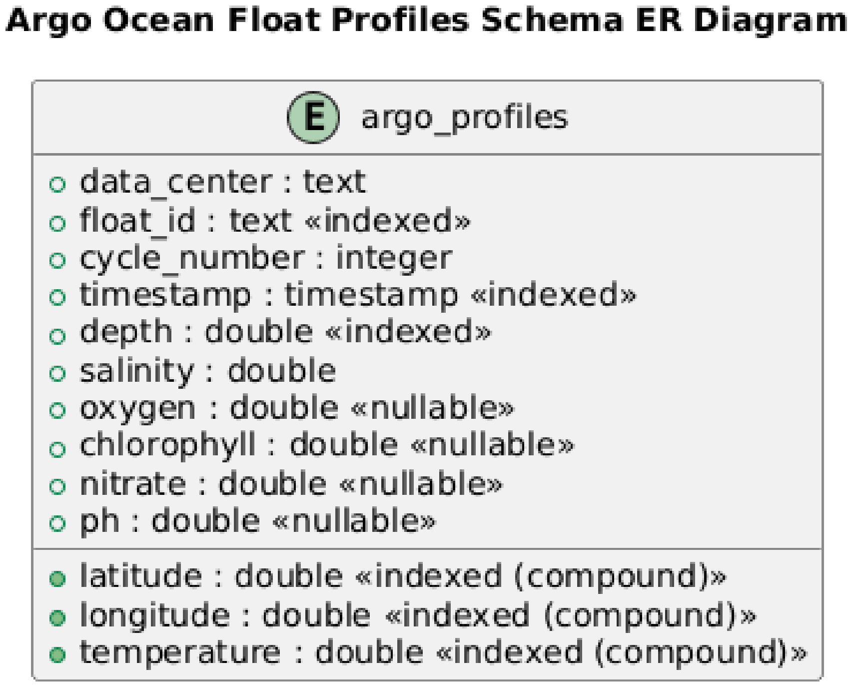


Figure 7.3: Database Schema Diagram

7.4 Backend API

The backend is implemented using a high-performance API framework that exposes multiple endpoints for data retrieval and analysis. These endpoints support functionalities such as accessing float positions, retrieving depth profiles, generating time-series data, and performing advanced computations like ocean heat content estimation.

The API is designed to handle concurrent requests efficiently and ensures secure communication through proper configuration and environment management. Special attention is given to handling numerical operations correctly, ensuring compatibility with high-precision database fields. This layer serves as the bridge between the data storage system and the user interface.

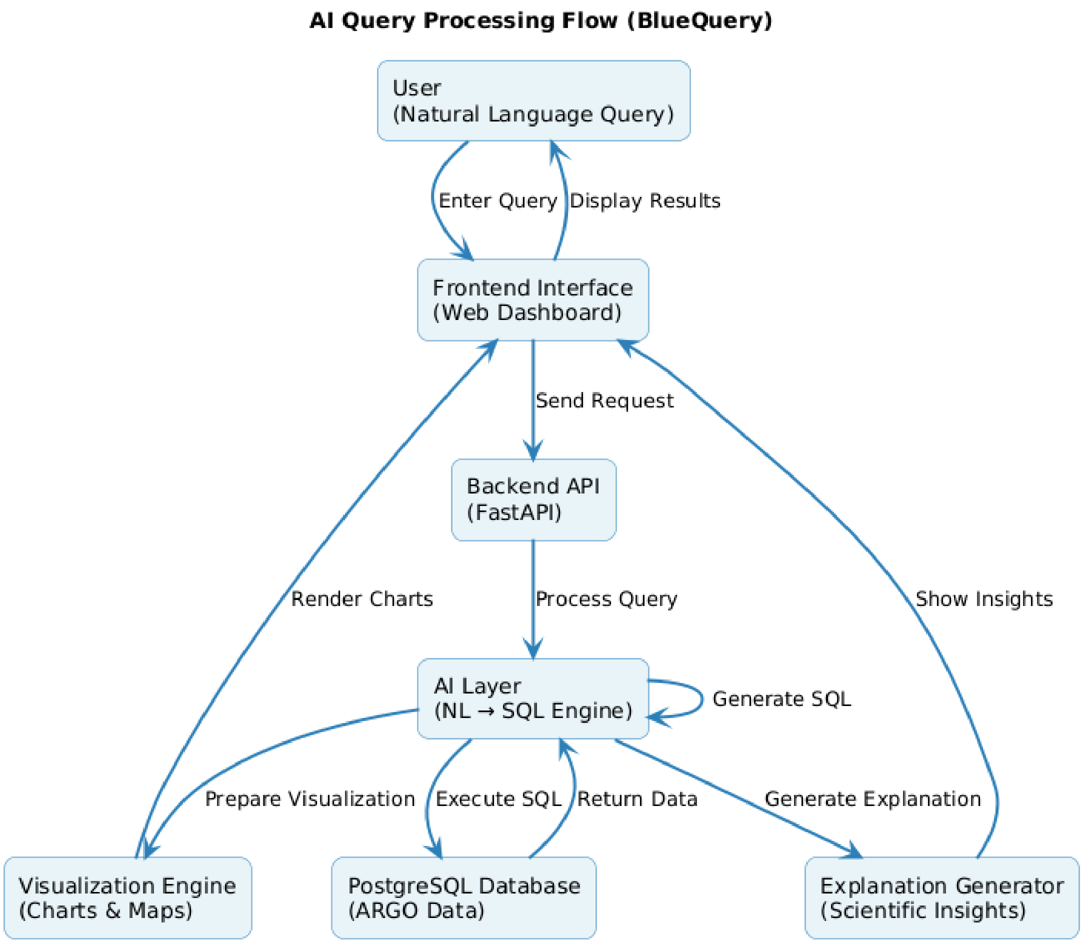


Figure 7.4: AI Query Processing Flow

7.5 AI Layer

The AI Layer enables natural language interaction by converting user queries into structured database queries. It utilizes a multi-model approach to ensure reliability, where fallback mechanisms are implemented to handle failures or ambiguities. This ensures that the system consistently generates accurate responses.

The AI system incorporates schema-aware query generation, reducing errors related to incorrect data references. Additional validation mechanisms are implemented to correct queries before execution. Beyond query translation, the AI layer performs analytical computations such as thermocline detection, ocean heat content calculation, and feature importance analysis. It also generates clear, human-readable explanations of results, making complex scientific outputs easier to understand.

7.6 Interface Layer

The Interface Layer provides a web-based platform for user interaction. It allows users to input queries in natural language and receive results in the form of structured data and visual outputs. The interface is designed to be intuitive, reducing the need for technical expertise.

Interactive visualizations such as charts and maps are used to represent oceanographic parameters effectively. These include depth profiles, time-series trends, and spatial distributions. The design focuses on clarity and responsiveness, enabling users to explore data efficiently without manual preprocessing or multiple tools.

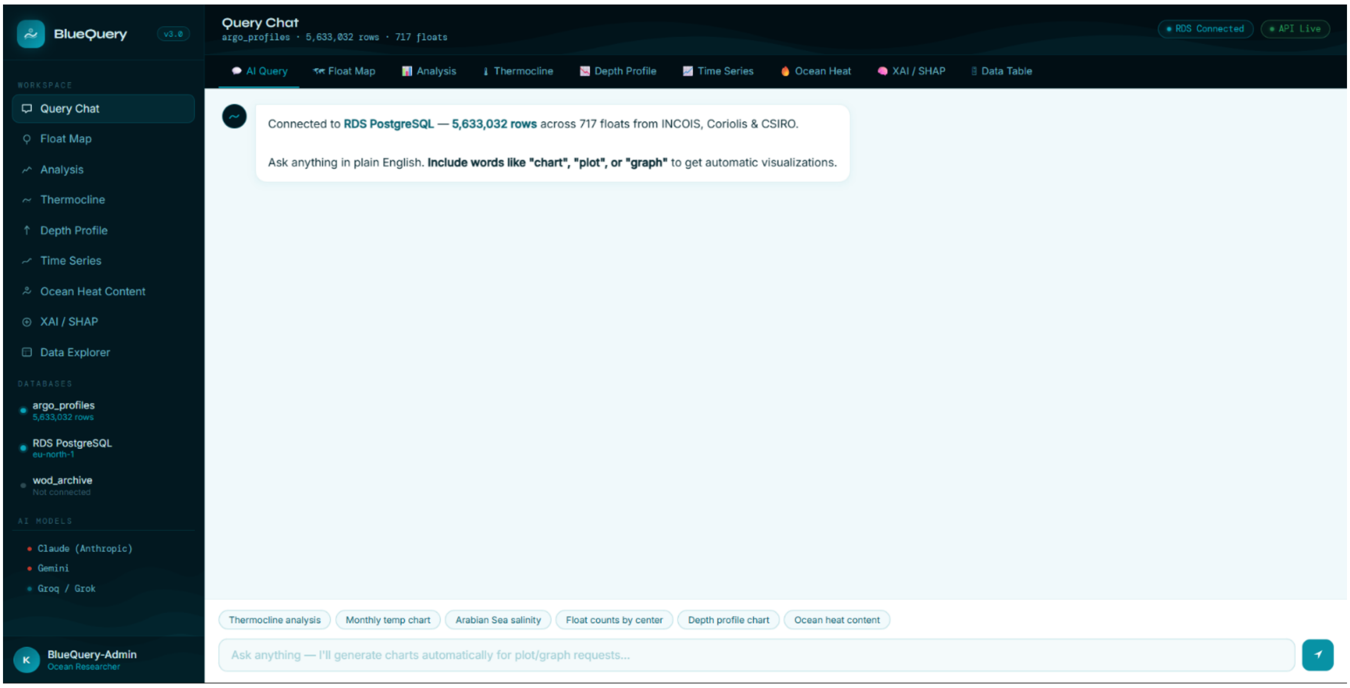


Figure 7.6: Dashboard Interface

7.7 Technology Stack

The system integrates a combination of cloud infrastructure, data processing tools, backend technologies, artificial intelligence models, and frontend visualization libraries. Cloud services are used for data storage and computation, while Python-based tools handle data extraction and transformation. The backend framework enables efficient API communication, and visualization libraries support interactive data representation.

Artificial intelligence components facilitate natural language querying and explanation generation, while numerical computation libraries support scientific analysis. This integrated technology stack ensures that the system is scalable, efficient, and capable of handling complex oceanographic data workflows.

CHAPTER 8
TESTING AND FEEDBACK

The Testing and Feedback stage focuses on evaluating the developed prototype to assess its performance, usability, effectiveness, and reliability in addressing the defined problem. This stage involves observing system behavior, collecting user feedback, and identifying areas for improvement.

The prototype was tested with different categories of users including students, researchers, educators, and non-technical users to understand system functionality and ease of interaction. Feedback was collected based on query handling, response time, visualization clarity, analytical usefulness, and overall user experience.

8.1 Testing Approach

The system was evaluated by allowing users to interact with the platform using natural language queries. Users explored multiple functionalities including data retrieval, visualization, thermocline detection, salinity analysis, ocean trend analysis, and time-series exploration to assess overall system performance.

The testing primarily focused on:

- Ease of interaction with the system
- Accuracy of query interpretation
- Speed of response generation
- Clarity of visual outputs
- Reliability of analytical results
- Usefulness of generated explanations

The system was also tested using different combinations of oceanographic parameters such as temperature, salinity, pressure/depth, timestamp, latitude, and longitude to evaluate data handling and analytical consistency.

8.2 Feedback Summary

The feedback obtained from users highlighted both the strengths of the system and areas requiring improvement.

Users reported that the system is easy to use, with an intuitive interface that allows interaction without technical expertise. The response time was observed to be fast, enabling efficient data exploration and reducing delays in obtaining results. Visualizations such as charts and maps were found to be clear and effective for understanding oceanographic patterns and trends.

Researchers appreciated the reduction in preprocessing effort compared to traditional workflows involving NetCDF parsing and separate analytical tools. The natural language interface was identified as a major advantage because it simplified interaction with large-scale oceanographic datasets.

Users also explored the potential of the system for broader ocean intelligence applications using available parameters such as temperature, salinity, pressure/depth, timestamp, latitude, and longitude. Suggestions included:

- Marine organism habitat analysis
- Fish migration pattern analysis
- Ship route and navigation support
- Environmental condition monitoring
- Ocean trend prediction
- Region-wise climate variation analysis

The spatial and temporal characteristics of ARGO data were recognized as useful for detecting ocean condition changes and supporting future predictive systems.

However, certain limitations were identified during testing. In some cases, complex queries were not interpreted correctly during the initial stages, leading to inaccurate or incomplete outputs. Users also requested more detailed explanations alongside scientific visualizations to improve interpretability for non-technical users.

The feedback collected during this stage played an important role in improving query interpretation, visualization clarity, explanation generation, and overall system usability during the redesign phase.

8.3 Feedback from External Users

CHAPTER 9
RE-DESIGN AND IMPLEMENTATION

The Re-design and Implementation stage focuses on refining the developed system based on the feedback obtained during the testing phase. The objective of this stage is to improve system performance, enhance usability, and address the limitations identified earlier. By incorporating user feedback, the system was optimized to provide more accurate results, better visualization, and improved overall interaction.

9.1 Improvements Made

Based on the issues identified during testing, several improvements were implemented to enhance the functionality and reliability of the system. The accuracy of natural language query interpretation was significantly improved to reduce errors and ensure that user queries are correctly translated into structured database operations. To further increase robustness, a fallback mechanism using multiple AI models was introduced, allowing the system to maintain performance even when one model fails or produces uncertain outputs.

In addition, the visualization components were enhanced to provide clearer and more intuitive graphical representations of oceanographic data. Charts and visual outputs were refined to improve readability and user understanding. An explanation layer was also incorporated into the system, enabling it to generate descriptive insights alongside results. This addition helps users interpret the data more effectively, especially those without strong technical or domain knowledge.

9.2 Final Outcome

After implementing the above improvements, the system evolved into a fully functional and efficient platform capable of addressing the challenges identified in earlier stages. The final system allows users to interact with oceanographic data using natural language queries, eliminating the need for programming or specialized tools. It generates meaningful scientific insights from the data and presents them through interactive visual outputs such as charts and maps.

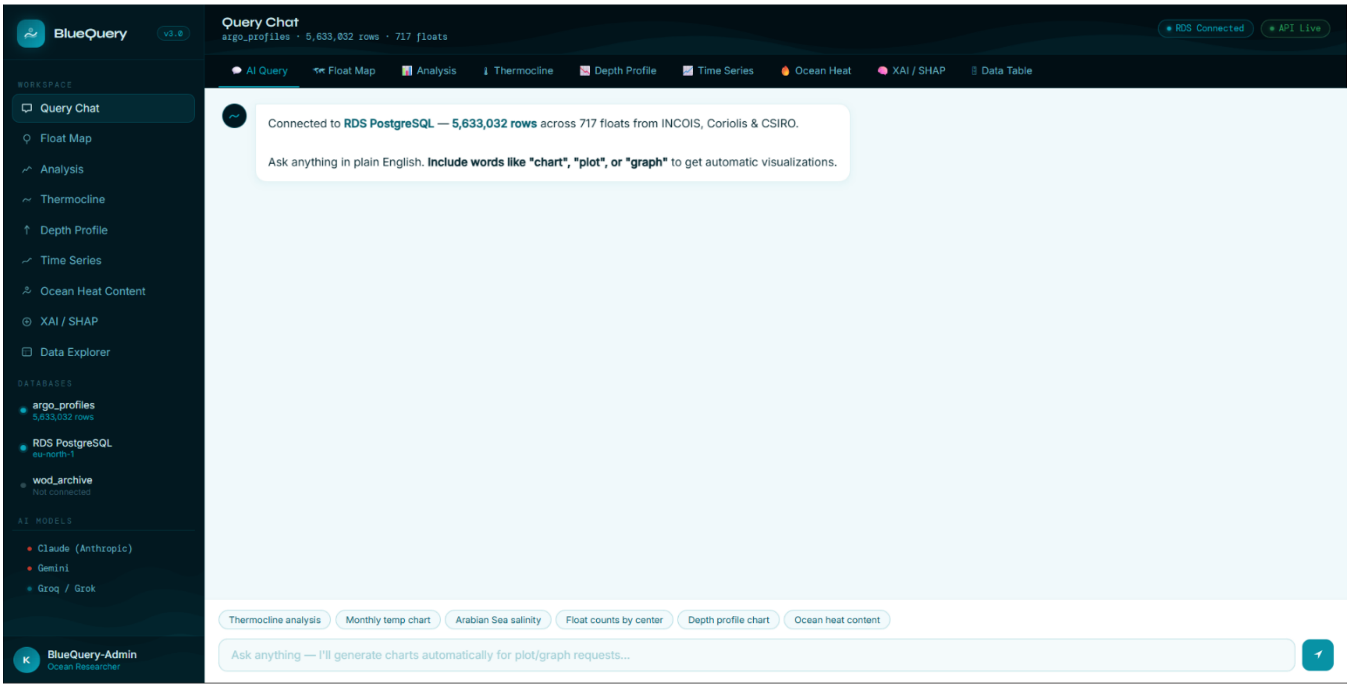
Overall, the refined system provides a seamless and user-friendly experience, combining data accessibility, analytical capability, and visualization in a single integrated platform.

CHAPTER 10
RESULT AND CONCLUSION

10.1 Results

BlueQuery successfully delivers on its core objective: making ARGO oceanographic data accessible to users regardless of technical background. The implemented system handles the full workflow from raw scientific data ingestion to user-level insight generation — automatically, in real time, through a natural language interface.

The platform ingests and serves 5.6 million depth-level observations from 717 ARGO floats spanning 2019 to 2026, sourced from INCOIS, Coriolis, and CSIRO. Users can query this dataset in plain English and receive SQL-generated results, interactive charts, and scientific explanations within seconds. Scientific analyses including thermocline detection, ocean heat content estimation, salinity profiling, and time-series trend analysis are available through both the conversational interface and dedicated analytical tabs.



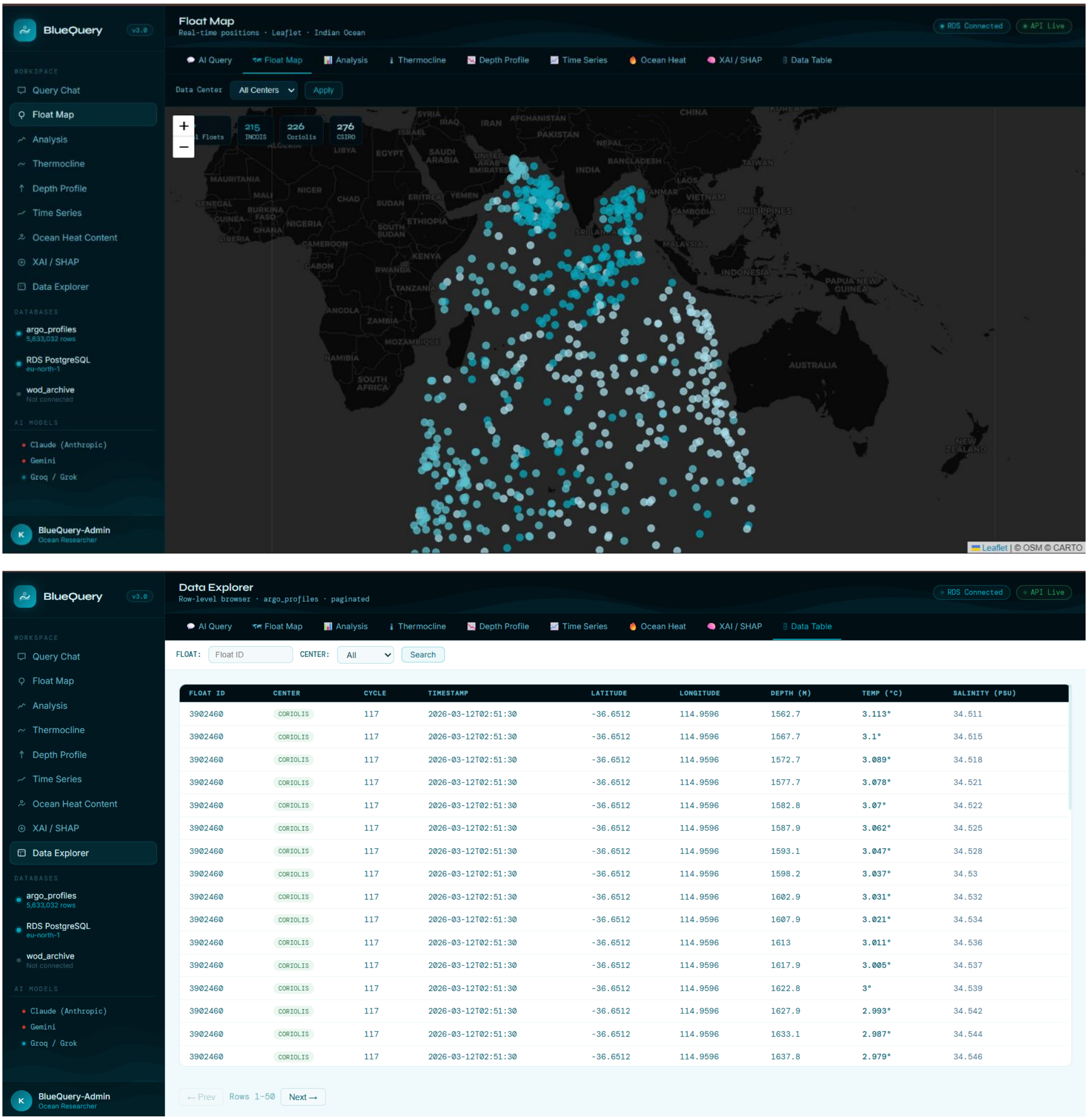


Figure 10.1 Sample System Output (Query + Chart + Explanation)

10.2 Conclusion

The project demonstrates that the gap between ocean data availability and ocean data usability is a design problem, not a data problem. A design thinking approach, applied rigorously from empathy through implementation, produced a system that eliminates the need for NetCDF parsing tools, programming environments, or domain expertise at the point of user interaction. A student, a policymaker, and a marine researcher can all extract meaningful insights from the same platform, calibrated to their own level of engagement and need.

The integration of data engineering, cloud infrastructure, large language models, and domain-specific scientific computation into a single cohesive user experience validates the design thinking methodology as applicable and effective in technically complex, data-intensive domains.

CHAPTER 11
FUTURE WORK

The following directions are proposed to extend the platform's capabilities and reach:

- Integration of satellite altimetry and sea surface temperature data to complement in-situ ARGO profiles with broader spatial coverage
- Real-time data streaming from GDAC feeds to ensure the database reflects the most current ARGO observations
- Expansion to BGC float parameters including dissolved oxygen, pH, and chlorophyll for marine ecosystem monitoring
- Development of a mobile application for field and operational use
- Incorporation of predictive analytics for sea surface temperature forecasting and cyclone risk assessment
- Anomaly detection pipeline to automatically flag unusual oceanographic events and generate alerts
- Extension of geographic coverage beyond the Indian Ocean to include Atlantic and Pacific ARGO data

CHAPTER 12
LEARNING OUTCOMES

12.1 Design Thinking in Practice

The project demonstrated that applying structured human-centered design methodology to a technical problem produces meaningfully different outcomes than purely engineering-driven development. Every architectural decision — from the conversational interface to the explanation layer — traces directly back to a specific user need identified in the empathy phase. The iterative loop between testing and re-design proved essential: the explanation layer and visualization refinements would not have been prioritized without user feedback exposing their absence.

12.2 Data Engineering at Scale

The ETL pipeline required solving real-world challenges: heterogeneous schema resolution across three international institutions, format conversion across 71,000 NetCDF files, quality flag interpretation, and cloud infrastructure configuration on AWS. These are professional-grade data engineering problems requiring judgment, not just execution.

12.3 AI Integration in Production Systems

Implementing a multi-LLM fallback engine with schema-aware prompting, automatic SQL validation, and structured intent detection reflects how AI is integrated into production systems in the industry. The project provided direct experience with prompt engineering, API orchestration across multiple providers, and systematic error handling — skills directly transferable to professional AI engineering roles.

12.4 User-Centered System Design

The most significant learning was the importance of designing for interpretation, not just for output. A system that returns correct data but presents it without context fails the user. The explanation layer, zone color-coding, and observation density annotations were all design decisions driven by this insight — and all emerged from the empathy and testing phases rather than from technical specification.

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